

# Muscle Contraction Without the Panic

This topic has a lot of new words, but only one story. If you can tell the story in plain English, the vocabulary will stick to it.

**The whole story in one sentence:** a nerve rings the muscle's doorbell, the alarm unlocks a vault of calcium, and calcium lets a tug-of-war team pull the rope.

## The three acts, as a story

| Act                                     | The story version                                                                                                                                                                                                                                           | The real version                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Act 1</b><br>Nerve talks to muscle   | A text message (action potential) arrives at the end of the nerve. The nerve rings the muscle's <b>doorbell</b> by squirting a chemical across a tiny gap. The door opens a crack and a wave spreads across the muscle like a stadium wave.                 | The action potential reaches the <b>axon terminal</b> . $\text{Ca}^{2+}$ enters, and vesicles release <b>ACh</b> into the <b>synaptic cleft</b> . ACh binds receptors on the <b>motor end plate</b> , $\text{Na}^+$ rushes in, and a muscle action potential spreads along the <b>sarcolemma</b> .                              |
| <b>Act 2</b><br>Signal releases calcium | The alarm runs down the <b>hallways</b> into the middle of the building. Next to each hallway is a <b>vault</b> full of calcium. The alarm unlocks the vault doors.                                                                                         | The action potential travels down the <b>T-tubules</b> . Voltage sensors (DHPR) pull open release channels (RyR) on the <b>sarcoplasmic reticulum (SR)</b> . $\text{Ca}^{2+}$ floods into the cytosol. This is <b>excitation-contraction coupling</b> .                                                                         |
| <b>Act 3</b><br>Filaments slide         | The <b>rope</b> has handholds, but a <b>cover</b> lies over them. Calcium is the <b>key</b> that opens the <b>lock</b> holding the cover. Now the team can <b>grab, pull, let go, and reach forward</b> , over and over. The ropes slide toward the middle. | $\text{Ca}^{2+}$ binds <b>troponin</b> , which moves <b>tropomyosin</b> off the binding sites on <b>actin</b> . <b>Myosin</b> heads bind (cross-bridge), pivot (power stroke), release when <b>ATP</b> binds, and re-cock when ATP is split. Thin filaments slide toward the <b>M line</b> , and the <b>sarcomere</b> shortens. |
| <b>Relaxation</b>                       | Stop ringing the doorbell. Lock the calcium back in the vault. Put the covers back over the handholds.                                                                                                                                                      | <b>AChE</b> breaks down ACh. SR pumps use <b>ATP</b> to move $\text{Ca}^{2+}$ back into the SR. Tropomyosin covers the binding sites again.                                                                                                                                                                                     |

## Where ATP shows up (students often miss the last two)

### 1. Re-cocking

Splitting ATP re-energizes the myosin head.

### 2. Letting go

ATP binding makes myosin release actin. No ATP = stuck (rigor mortis).

### 3. Relaxing

SR pumps use ATP to put  $\text{Ca}^{2+}$  back in storage.

## When a question scares you, ask these three things

1. Which act am I in?
2. What is this part's one job?
3. If that job fails, what is the next step that can't happen?

Almost every "what if" question on this topic can be solved by finding the broken link and following the chain forward.

## Fill in the chain

Use the word bank. Some words are used more than once. Say each line out loud after you fill it in.

**Word bank:** ACh ATP  $\text{Ca}^{2+}$   $\text{Na}^+$  T-tubules sarcoplasmic reticulum troponin tropomyosin  
actin myosin M line AChE

- 1 An action potential arrives at the axon terminal, and \_\_\_\_\_ enters the terminal.
- 2 Vesicles release \_\_\_\_\_ into the synaptic cleft.
- 3 It binds receptors on the motor end plate, and \_\_\_\_\_ rushes into the fiber.
- 4 A muscle action potential spreads along the sarcolemma and down the \_\_\_\_\_.
- 5 \_\_\_\_\_ is released from the \_\_\_\_\_.
- 6  $\text{Ca}^{2+}$  binds \_\_\_\_\_, which pulls \_\_\_\_\_ off the binding sites.
- 7 \_\_\_\_\_ heads bind to \_\_\_\_\_, forming cross-bridges.
- 8 The power stroke pulls the thin filaments toward the \_\_\_\_\_.
- 9 \_\_\_\_\_ binds myosin so it lets go, and is then split to re-cock the head.
- 10 To relax, \_\_\_\_\_ breaks down ACh and pumps return \_\_\_\_\_ to the SR.

### Worked examples: how to think it through

#### QUESTION

**A drug blocks the ACh receptors on the motor end plate. What happens to contraction?**

#### WHICH ACT?

ACh receptors are on the muscle side of the junction. That's Act 1.

#### WHAT IS THE PART'S JOB?

When ACh binds, the receptor opens and lets  $\text{Na}^+$  in. That starts the muscle action potential.

#### WHAT CAN'T HAPPEN NEXT?

No  $\text{Na}^+$  in → no muscle action potential → nothing goes down the T-tubules → no  $\text{Ca}^{2+}$  released → myosin can't bind. **The muscle can't contract. It stays limp.**

#### QUESTION

**During contraction, which bands of the sarcomere get shorter, and which stay the same?**

#### WHICH ACT?

Sliding filaments, Act 3.

#### KEY FACT

The filaments never change length. They only overlap more.

#### REASON IT OUT

The A band is the length of the thick filaments, so it stays the same. The I band (thin only) and H zone (thick only) shrink because the overlap grows. The Z discs move closer, so the sarcomere shortens.

## Your turn

Use the three questions: **Which act? What's the part's job? What can't happen next?** Write in full sentences. There is no penalty for getting these wrong. They are practice.

1. Where is the  $\text{Ca}^{2+}$  that triggers contraction stored in a skeletal muscle fiber, and what makes it come out?

---

---

---

2. In a resting muscle, myosin and actin are right next to each other. Why don't they bind?

---

---

---

3. A few hours after death, the body becomes stiff (rigor mortis). Use the cross-bridge cycle to explain why.

---

---

---

4. Some pesticides block acetylcholinesterase (AChE). Predict what happens to the muscles and explain why.

---

---

---

5. Relaxation is not just "turning off." Name two places relaxation (or getting ready to relax) needs ATP.

---

---

---

6. Bonus: Explain the whole contraction to a friend in five sentences or fewer, without using the word "basically."

---

---

---

---

## Answer key

### Fill in the chain

1  $\text{Ca}^{2+}$  · 2 ACh · 3  $\text{Na}^+$  · 4 T-tubules · 5  $\text{Ca}^{2+}$ , sarcoplasmic reticulum · 6 troponin, tropomyosin · 7 Myosin, actin · 8 M line · 9 ATP · 10 AChE,  $\text{Ca}^{2+}$

### Your turn

1. In the sarcoplasmic reticulum (SR). The muscle action potential travels down the T-tubules. Voltage sensors (DHPR) in the T-tubule change shape and pull open release channels (RyR) on the SR, so  $\text{Ca}^{2+}$  flows out into the cytosol.
2. Tropomyosin lies over the myosin-binding sites on actin. Until  $\text{Ca}^{2+}$  binds troponin and troponin moves tropomyosin, myosin has nothing to grab.
3. After death, the cells run out of ATP. Myosin needs ATP to let go of actin. Calcium also leaks from the SR and can't be pumped back, so heads bind actin but can't release. The cross-bridges lock, and the muscles stay stiff.
4. ACh is never broken down, so it keeps binding receptors and keeps starting action potentials. Muscles twitch, spasm and have trouble relaxing, including the breathing muscles. Broken link: the first step of relaxation.
5. Any two of: SR pumps (SERCA) use ATP to move  $\text{Ca}^{2+}$  back into the SR; ATP binding lets myosin release actin; and the  $\text{Na}^+/\text{K}^+$  pump restores the membrane after action potentials.
6. Sample: The motor neuron releases ACh, which makes an action potential spread across the muscle fiber. The action potential goes down the T-tubules and makes the SR release calcium. Calcium binds troponin, which moves tropomyosin off actin's binding sites. Myosin heads then grab actin, pull it toward the middle, let go using ATP, and repeat. The thin filaments slide over the thick ones, so the sarcomere and the whole muscle shorten.

Want to see it move? Open the **Contraction Walkthrough** and go one step at a time. Then try the **Contraction Chain** game.